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BRANCH: CSM
SECTION: B
COURSE NAME: FULL STACK WEB DEVELOPMENT
CODE: 23CM4121

TITLE OF THE EXPERIMENT

To create a JavaScript-based Student Profile Application using classes, objects, DOM manipulation, dynamic HTML elements, and event handling.

AIM

To develop a Student Profile Application that accepts student details and dynamically displays them using JavaScript classes and DOM manipulation.

DESCRIPTION

The Student Profile Application is a simple web application developed using HTML, CSS, and JavaScript. It uses a JavaScript Student class with properties such as Name, Roll Number, Department, and CGPA. The user enters the student details through input fields and clicks the Display Student Profile button. JavaScript collects the entered values using DOM methods and creates a Student object using the class constructor. The program dynamically creates HTML elements using "createElement()", inserts data using "textContent", and displays the generated profile using "appendChild()". Event handling is implemented using the "addEventListener()" method.

PROGRAM

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">


  <title>Student Profile</title>
```

```
<style>
```

```
body{  
    font-family: Arial, sans-serif;  
    background-color: #f2f2f2;  
    text-align: center;  
    padding: 30px;  
}
```

```
.container{  
    width: 400px;  
    margin: auto;  
    background-color: white;  
    padding: 25px;  
    border-radius: 10px;  
    box-shadow: 0 0 10px gray;  
}
```

```
input{  
    width: 90%;  
    padding: 10px;  
    margin: 8px;  
    border: 1px solid #ccc;  
    border-radius: 5px;  
}
```

```
button{  
    padding: 10px 20px;  
    background-color: #007bff;
```

```
    color: white;

    border: none;

    border-radius: 5px;

    cursor: pointer;

    margin-top: 10px;
}
```

```
button:hover {

    background-color: #0056b3;
}
```

```
.profile {

    margin-top: 25px;

    padding: 20px;

    background-color: #e8f4ff;

    border-radius: 10px;

    text-align: left;

    border-left: 5px solid #007bff;
}
```

```
;
```

```
.profile h2 {

    text-align: center;

    color: #007bff;
}
```

```
.profile p {

    font-size: 16px;

    margin: 10px 0;
```

```
    }
</style>
</head>

<body>

    <div class="container">

        <h1>Student Profile</h1>

        <input type="text" id="name" placeholder="Enter Name">

        <input type="text" id="roll" placeholder="Enter Roll Number">

        <input type="text" id="department" placeholder="Enter Department">

        <input type="number" id="cgpa" step="0.01" placeholder="Enter CGPA">

        <br>

        <button id="displayBtn">Display Student Profile</button>

        <div id="profileContainer"></div>

    </div>

    <script>
```

```
// Creating Student class
```

```
class Student {
```

```
    constructor(name, rollNumber, department, cgpa) {
```

```
        this.name = name;
```

```
        this.rollNumber = rollNumber;
```

```
        this.department = department;
```

```
        this.cgpa = cgpa;
```

```
    }
```

```
}
```

```
// Selecting the button
```

```
const button = document.getElementById("displayBtn");
```

```
// Adding click event
```

```
button.addEventListener("click", function() {
```

```
    // Getting values from input fields
```

```
    const name = document.getElementById("name").value;
```

```
    const rollNumber = document.getElementById("roll").value;
```

```
    const department = document.getElementById("department").value;
```

```
    const cgpa = document.getElementById("cgpa").value;
```

```
// Creating Student object
```

```
const student = new Student(
```

```
    name,
```

```
    rollNumber,
```

```
        department,
        cgpa
    );

// Selecting profile container
const profileContainer =
    document.getElementById("profileContainer");

// Clear previous profile
profileContainer.innerHTML = "";

// Creating profile div dynamically
const profile = document.createElement("div");

profile.className = "profile";

// Creating heading
const heading = document.createElement("h2");

heading.textContent = "Student Profile";

// Creating paragraphs
const namePara = document.createElement("p");

namePara.textContent = "Name    : " + student.name;

const rollPara = document.createElement("p");
```

```
rollPara.textContent = "Roll No  : " + student.rollNumber;
```

```
const departmentPara = document.createElement("p");
```

```
departmentPara.textContent =
```

```
    "Department : " + student.department;
```

```
const cgpaPara = document.createElement("p");
```

```
cgpaPara.textContent = "CGPA    : " + student.cgpa;
```

```
// Adding elements to profile
```

```
profile.appendChild(heading);
```

```
profile.appendChild(namePara);
```

```
profile.appendChild(rollPara);
```

```
profile.appendChild(departmentPara);
```

```
profile.appendChild(cgpaPara);
```

```
// Adding profile to webpage
```

```
profileContainer.appendChild(profile);
```

```
});
```

```
</script>
```

</body>

</html>

OUTPUT

The screenshot displays a web application interface for a student profile. It features a form with four input fields: Name (mohini), Roll No (A24126552110), Department (csm), and CGPA (9.0). A blue button labeled 'Display Student Profile' is positioned below the form. Below the button, a light blue box contains the displayed student profile information.

Student Profile

mohini

A24126552110

csm

9.0

Display Student Profile

Student Profile

Name : mohini

Roll No : A24126552110

Department : csm

CGPA : 9.0